# STA347 — Probability

Course Notes

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## Lecture 1: Set Theory Fundamentals

Sep 8, 2026

"Probability = Set Theory + Measure"

1. **Basic Set Operations**
   Union, Intersect, Complement, ...
   $A \cup B,\ A \cap B,\ A^C, \ldots$
2. **Def 1.1.1 Convergence of Real-Valued Sequences**

A sequence $(x_n)_{n=1}^{\infty} \subseteq \mathbb{R}$ converges to $x$ if

$$\forall \varepsilon > 0,\ \exists N \in \mathbb{N} \text{ s.t. } n > N \Rightarrow |x_n - x| < \varepsilon.$$

$x_n \to x$ converge to $x$?
Pick $\varepsilon$ (how close $x_n$ is to the target).
For any $\varepsilon$ picked, can find $N$, such that $n > N$, $|x_n - x| < \varepsilon$.
e.g. $x_n = \dfrac{1}{n} \to 0$

3. **Def 1.1.2 Functional Convergence Pointwisely**

   A sequence $(f_n)_{n=1}^{\infty} \subseteq \mathbb{R}^X$ converges pointwise to $f$ (target $f$) if

   $$\forall \varepsilon > 0,\ \forall x \in X,\ \exists N_x \in \mathbb{N} \text{ s.t. } n > N_x \Rightarrow |f_n(x) - f(x)| < \varepsilon.$$

   $f_1, \ldots, f_n, \ldots$ — sequence of functions
   $f$ — target function
   For any $\varepsilon$ and $x$, can find $N_x$ s.t. $n > N_x$, $|f_n(x) - f(x)| < \varepsilon$.
   $f_1(2), f_2(2), \ldots, f_n(2), \cdots \to f(2)$ — for every $x$, there is a corresponding $N_x$.

4. **Def 1.1.3 Functional Convergence Uniformly**

   A sequence $(f_n)_{n=1}^{\infty} \subseteq \mathbb{R}^X$ converges uniformly to $f$ if

   $$\forall \varepsilon > 0,\ \exists N \in \mathbb{N} \quad \forall x \in X,\ n > N \Rightarrow |f_n(x) - f(x)| < \varepsilon.$$

   That works for all $x \in X$ — need a common threshold $N$ for uniform convergence.

**Key difference:**

For pointwise convergence, you need to have $x$ specified first, then there is a threshold $N_x$ corresponding to each $x$.
For uniform convergence, you need a common threshold $N$ that works for all $x \in X$.

**Example 1**

$$f_n(x) = \frac{x}{n + x}, \quad x \geq 0$$

Then for any fixed value $x$, as $n \to \infty$, $f_n(x) \to 0$:
$x = 1$: $f_n(1) = \dfrac{1}{n+1} \to 0$ as $n \to \infty$.
$x = 2$: $f_n(2) = \dfrac{2}{n+2} \to 0$ as $n \to \infty$.

$$\therefore f_n(x) \xrightarrow{\text{pointwise}} 0$$

However, $f_n$ does not converge uniformly to 0: no $N$ works for all $x$.
No matter how you specify $N$, there is always a larger value of $x$ for which the bound fails.

$$\Rightarrow f_n(x) \not\xrightarrow{\text{uniformly}} 0$$

## Limiting Processes

### Def 1.2.1 The Supremum of a Set

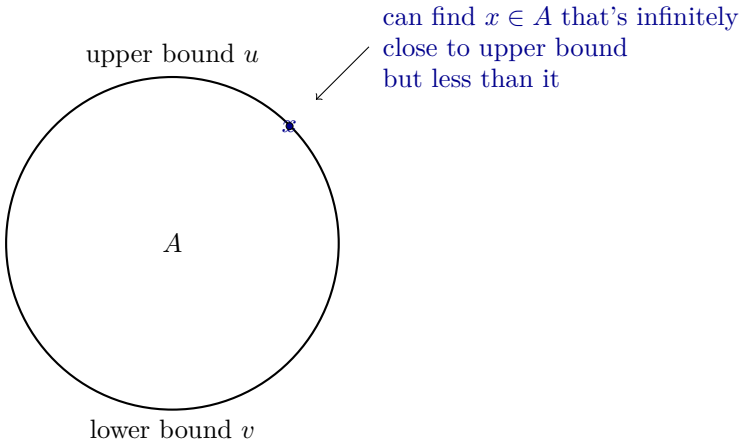
Let $A \subseteq \mathbb{R}$ be a subset. $u = \sup A$ iff

$$\forall \varepsilon > 0,\ \exists x \in A \quad u - x < \varepsilon \quad \text{and} \quad \forall x \in A,\ x \leq u.$$

### Def 1.2.2 The Infimum of a Set

Let $A \subseteq \mathbb{R}$ be a subset. $v = \inf A$ iff

$$\forall \varepsilon > 0,\ \exists x \in A \quad x - v < \varepsilon \quad \text{and} \quad \forall x \in A,\ x \geq v.$$




**Example 2**

$$A = (0, 1)$$

$$\sup A = 1 \qquad \inf A = 0$$

### Distinction of sup/inf vs. max/min:

$\max(A)$ and $\min(A)$ must be elements of the set $A$.
$\sup(A)$ and $\inf(A)$ may not be elements of $A$.

**Example 3**

$$A = (0, 1)$$

$\max(A)$ and $\min(A)$ do not exist. $\sup(A)$ and $\inf(A)$ do exist.

$$B = (0, 1]$$

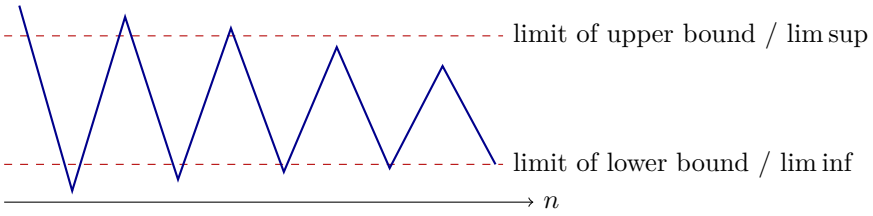
$\min(B)$ does not exist, $\max(B) = 1$. $\sup(B) = 1$, $\inf(B) = 0$.

### Def 1.2.3 Limsup and Liminf

Let $(x_n) \subseteq \mathbb{R}$ be a sequence.

$$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} \sup_{m\geq n} x_m = \inf_{n\geq 0} \sup_{m\geq n} x_m$$

$$\liminf_{n\to\infty} x_n = \lim_{n\to\infty} \inf_{m\geq n} x_m = \sup_{n\geq 0} \inf_{m\geq n} x_m$$




**Example 4**

$$x_n = (-1)^n = -1, 1, -1, 1, \ldots$$

No limit, diverges: $\lim_{n\to\infty} x_n$ DNE.
But $\sup_{m\geq n} x_m = 1$ and $\inf_{m\geq n} x_m = -1$

$$\Rightarrow \limsup_{n\to\infty} x_n = 1, \qquad \liminf_{n\to\infty} x_n = -1$$

The limsup and liminf are used to describe the long-run behaviour of a sequence, even if the sequence does not c
For a sequence $x_n$, if $\limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n = x$, then $\lim_{n\to\infty} x_n = x$.

## Function Limits

### Def 1.3 Left and Right Limit of a Function

For any function $f : \mathbb{R} \to \mathbb{R}$,

$$f(x-) = \lim_{y\uparrow x} f(y) = L \quad \text{(to the left of } x\text{, a small neighbourhood)}$$

$$\Leftrightarrow \forall \varepsilon > 0,\ \exists \delta > 0 \text{ s.t. } \forall y \in (x-\delta, x),\ |f(y) - L| < \varepsilon.$$

$$f(x+) = \lim_{y\downarrow x} f(y) = U$$

$$\Leftrightarrow \forall \varepsilon > 0,\ \exists \delta > 0 \text{ s.t. } \forall y \in (x, x+\delta),\ |f(y) - U| < \varepsilon.$$

If $f(x-) = f(x+) = C$, then $\lim_{y\to x} f(y) = C$.

**Example 5**

$$f(x) = \begin{cases} 0 & x < 0 \\ 1 & x \geq 0 \end{cases}$$

$f(0-) = 0,\ f(0+) = 1$

$$\Rightarrow \lim_{x\to 0} f(x) \text{ DNE}$$

# Lecture 2: Probability Measure

Sep 10, 2026

### Def 2.1 Sample Space

A sample space $\Omega$ is any non-empty set.

**Example 2.1** The sample space contains "all the states of the following"
Coin Flipping: $\Omega = \{H, T\}$
Weather: {hot, cold, mild} $\times$ {wet, dry}

### Def 2.2 Probability Measure (STA347 ver.)

For a sample space $\Omega$, we say $P$ is a probability measure if the following holds:

i) $\forall E \subseteq \Omega, 0 \leq P(E) \leq 1$, that is $P : \mathcal{P}(\Omega) \to [0, 1]$
ii) $P(\Omega) = 1$
iii) $\forall E_1, E_2, \cdots \subseteq \Omega$ such that $E_i \cap E_j = \varnothing \quad \forall\, i \neq j$

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

$\mathcal{P}(\Omega)$ is the power set of $\Omega$.

**Remark:** It may be impossible for such a $P$ to exist. To define exactly which subsets we want $P$ to satisfy these axioms, that would require more advanced mathematical theories. For this class, we just assume such a $P$ is defined for all subsets of $\Omega$.

**Example 2.2 Probabilities of Coin Flipping**

$$\Omega = \{H, T\}$$
$$\mathcal{P}(\Omega) = \{\phi, \{H\}, \{T\}, \{H, T\}\}$$
$$P : \mathcal{P}(\Omega) \mapsto [0, 1]$$

$$P : \phi \mapsto 0$$
$$\{H\} \mapsto \frac{1}{2}$$
$$\{T\} \mapsto \frac{1}{2}$$
$$\{H, T\} \mapsto 1$$

**Example 2.3 The Uniform/Lebesgue Measure**
Let $\Omega = [L, U]$ for some $L < U \in \mathbb{R}$. Define

$$P([a, b]) = \frac{b - a}{U - L} \quad \text{when } L \leq a \leq b \leq U$$

We assume such a measure exists (and is well defined) for this class.

**Proposition 2.3** The following properties are satisfied by any probability measure $P$

1. $\forall E \subseteq \Omega, P(E^C) = 1 - P(E)$
2. $P(\phi) = 0$
3. $\forall E, F \subseteq \Omega$, such that $E \subseteq F, P(E) \leq P(F)$ ("big event, big probability")
4. $\forall E, F \subseteq \Omega, P(E \bigcup F) = P(E) + P(F) - P(E \bigcap F)$
5. $\forall E, F \subseteq \Omega, P(E \bigcap F) = P(E) + P(F) - P(E \bigcup F)$
6. $\forall E, F \subseteq \Omega, P(F \setminus E) = P(F) - P(E \bigcap F)$

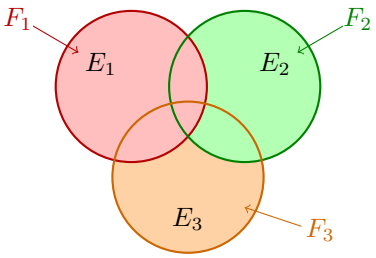
Proof via Def 2.2 (probability axioms).

**Lemma 2.4** For a sequence $E_1, \ldots, E_n \subseteq \Omega$

$$P(\bigcup_{i=1}^{n} E_i) \leq \sum_{i=1}^{n} P(E_i)$$

*Proof.* We define $F_i = E_i \setminus \bigcup_{j=1}^{i-1} E_j, F_1 = E_1$

$$\bigcup_{i=1}^{n} E_i = \bigcup_{i=1}^{n} F_i \text{ and } F_i \bigcap F_j = \phi \text{ if } i \neq j$$




$F_i$ = the part of $E_i$ not already covered by $E_1, \ldots, E_{i-1}$.

from the probability axiom.

$$\text{Since } \bigcup_{i=1}^{n} E_i = \bigcup_{i=1}^{n} F_i$$

$$P(\bigcup_{i=1}^{n} E_i) = P(\bigcup_{i=1}^{n} F_i)$$

$$= \sum_{i=1}^{n} P(F_i)$$

$$\text{We also know } F_i \subseteq E_i \text{ as } F_i = E_i \setminus \bigcup_{j=1}^{i-1} E_j$$

$$P(F_i) \leq P(E_i)$$

$$P(\bigcup_{i=1}^{n} E_i) = \sum_{i=1}^{n} P(F_i) \leq \sum_{i=1}^{n} P(E_i)$$

□

**Lemma 2.5** Let $P$ be the Lebesgue measure on $[0, 1]$. Then $P(E) = 0$ for any countable set $E$.

# Lecture 3: Independence, Random Variables and Distributions

Sep 15, 2026

### Def 2.7 Independence

Events $E_1, \ldots, E_n \subseteq \Omega$ are independent with respect to a probability measure $P$, if for each $\mathcal{I} \subseteq \underbrace{[n]}_{\text{any subset of}\{1,\ldots,n\}}$

$$P\left(\bigcap_{i\in\mathcal{I}} E_i\right) = \prod_{i\in\mathcal{I}} P(E_i)$$

e.g. if $\mathcal{I} = \{1,2,3\}$ a subset of $\{1,\ldots,n\}$

$$P(E_1 \bigcap E_2 \bigcap E_3) = P(E_1)P(E_2)P(E_3)$$

**Proposition 2.7.1** If $E_1, \ldots E_n$ are independent, then $E_1^c, \ldots, E_n^c$ are also independent. Proof as an exercise.

**Proposition 2.7.2** If an infinite collection of events $\{E_\alpha : \alpha \in \mathcal{I}\}$ are independent, then for a finite subset $\mathcal{J} \subseteq \mathcal{I}$, the events $\{E_\beta : \beta \in \mathcal{J}\}$ are independent.

## Random Variables and Distributions

### Def. 3.1 Random Variables

Given a sample space $\Omega$, a function $X : \Omega \to \mathbb{R}$ is a random variable.

### Example 3.1.1 Coin Flipping

$$\Omega = \{H, T\}$$

$$\downarrow \text{R.V. } X: \quad X(H) \to 1, \quad X(T) \to 0$$

$$\mathbb{R}$$

### Theorem 3.2 Distribution

A random variable $X$ and probability measure $P$ induce a probability measure $\mu$ on $\mathbb{R}$ defined by

$$\mu(A) = P(X \in A) = P(\{\omega \in \Omega : X(\omega) \in A\})$$

for any $A \subseteq \mathbb{R}$. $\mu$ is a measure on the real number set: $\mu(\{0,1\})$ and $\mu([0,1))$ each return a certain value. $\mu(A)$ takes a real number set $A$ as input, and equals $P$ of the pre-image of $A$ w.r.t. r.v. $X$, i.e. $P(\Omega_A) = \mu(A)$ where $\Omega_A = \{\omega \in \Omega : X(\omega) \in A\}$.

**Example 3.2.1** Define the following

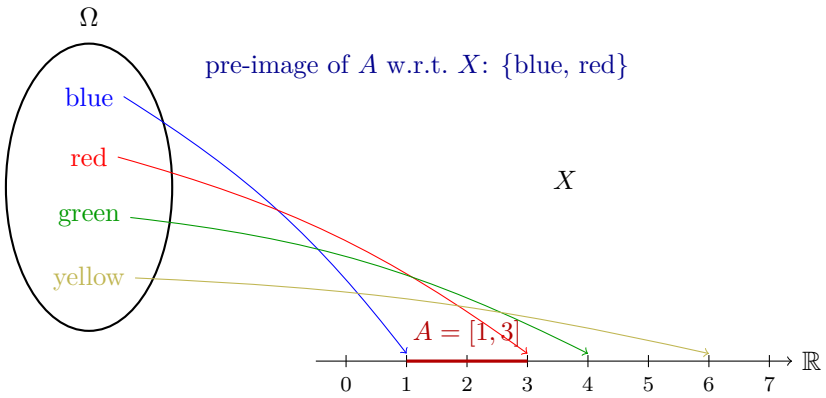
$$\Omega = \{\text{blue, red, green, yellow}\}$$

$$X : \begin{cases} \text{blue} \to 1 \\ \text{red} \to 3 \\ \text{green} \to 4 \\ \text{yellow} \to 6 \end{cases}$$

$$P : \{\text{blue}\} \mapsto \tfrac{1}{5}, \quad \{\text{red}\} \mapsto \tfrac{1}{5}, \quad \{\text{green}\} \mapsto \tfrac{1}{4}, \quad \{\text{yellow}\} \mapsto 1 - \tfrac{1}{5} - \tfrac{1}{5} - \tfrac{1}{4} = \tfrac{7}{20}$$

$$P(\{\text{blue, red}\}) = \tfrac{1}{5} + \tfrac{1}{5} = \tfrac{2}{5}$$

$$A = [1, 3]$$




1. $\{w : w \in \Omega, X(w) \in A\} = \{\text{blue ,red}\}$
2. $P(\{\text{blue ,red}\}) = \frac{2}{5}$, so $\mu(A) = \frac{2}{5}$

in summary

$$\mu([1, 3]) = \frac{2}{5}$$

**Claim:** $\mu$ is a probability measure on $\mathbb{R}$ (for any real number set $B$).

*Proof.* 1. $\mu(B) \in [0, 1]$ is trivial, since $P$ is a probability measure.

2. Since $X(\omega) \in \mathbb{R}$ for all $\omega \in \Omega$.

$$\mu(\mathbb{R}) = P(\{\omega : \omega \in \Omega, X(\omega) \in \mathbb{R}\}) = P(\Omega) = 1$$

3. Consider disjoint sets $A_1, A_2, \cdots \subseteq \mathbb{R}$ and define $E_i = \{\omega : X(\omega) \in A_i\}$
   (e.g. $E_1 = \{\omega : X(\omega) \in A_1\}$, $E_2 = \{\omega : X(\omega) \in A_2\}$)
   Since $X$ is a function, it is impossible for $X(\omega) = a$ and $X(\omega) = b$ when $a \neq b$, so the $E_i$'s

are also disjoint. Then

$$\begin{aligned}
\mu\left(\bigcup_{i=1}^{\infty} A_i\right) &= P\left(X(\omega) \in \bigcup_{i=1}^{\infty} A_i\right) \\
&= P\left(\bigcup_{i=1}^{\infty} E_i\right) \quad \text{we know } E_i\text{'s are disjoint} \\
&= \sum_{i=1}^{\infty} P(E_i) \\
&= \sum_{i=1}^{\infty} \mu(A_i)
\end{aligned}$$

□

**Def. 3.3 The Cumulative Distribution Function**
A random variable $X$ and measure $P$ generate the cumulative distribution function (CDF) defined by

$$F(x) = \mu((-\infty, x]) = P(X \leq x)$$

where $P(X \leq x)$ is a short-hand notation for

$$P(\{\omega : \omega \in \Omega, X(\omega) \in (-\infty, x]\})$$

**Theorem 3.3.1** A distribution function $F$ for a random variable $X$ uniquely defines the measure $\mu$.
Proof is outside the scope of this class.

**Example 3.3.2 Uniform Distribution**
$P$ is uniform on $[0, 1]$ and $X(\omega) = \omega$
For $x \in [0, 1)$

$$\begin{aligned}
F(x) &= \mu((-\infty, x]) && \leftarrow \text{by definition of CDF} \\
&= P(X \leq x) && \leftarrow \text{by Thm 3.2 (definition of distribution)} \\
&= P\left(\{\omega : X(\omega) \in (-\infty, x]\}\right) \\
&= P((-\infty, x]) && \leftarrow \text{by } X(\omega) = \omega \\
&= P([0, x]) && \leftarrow \text{because } P \text{ is a uniform/Lebesgue measure} \\
&= x
\end{aligned}$$

**Theorem 3.4 Properties of Distribution Function**
A distribution function satisfies the following properties:

i) For all $x \leq y \in \mathbb{R}, F(x) \leq F(y)$
ii) $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to +\infty} F(x) = 1$
iii) $F$ is right-continuous

*Proof.* i) $x \leq y \in \mathbb{R}$

$$F(x) = \mu((-\infty, x]), F(y) = \mu((-\infty, y])$$

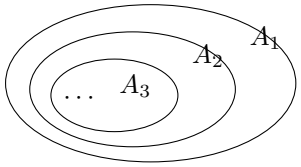
Since $x \leq y, \quad (-\infty, x] \subseteq (-\infty, y]$

$$\mu((-\infty, x]) \leq \mu((-\infty, y])$$

"bigger set, bigger probability"

$$(A \subseteq B, \quad P(A) \leq P(B) \quad / \quad \mu(A) \leq \mu(B))$$

ii) Define $A_n = \{X \leq -n\}, \quad n \geq 0$
Then $A_1 \supseteq A_2 \supseteq A_3 \supseteq \dots$

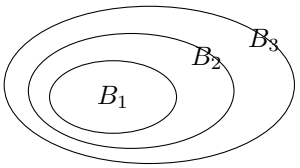



$\bigcap_{n=1}^{\infty} A_n = \phi$ (shrinks to nothing)

$$\bigcap_{n=1}^{\infty} A_n = \phi$$

$$\begin{aligned} \lim_{n\to\infty} F(-n) &= \lim_{n\to\infty} P(A_n) \\ &= P\left(\bigcap_{n=1}^{\infty} A_n\right) \\ &= P(\phi) = 0 \end{aligned}$$

Define $B_n = \{X \leq n\}$,
then $B_1 \subseteq B_2 \subseteq \dots$




$\cdots \quad \bigcup_{n=1}^{\infty} B_n = \Omega$ (grows to everything)

$$\bigcup_{n=1}^{\infty} B_n = \Omega$$

$$\begin{aligned} \lim_{n\to\infty} F(n) &= \lim_{n\to\infty} P(B_n) \\ &= P\left(\bigcup_{n=1}^{\infty} B_n\right) \\ &= P(\Omega) = 1 \end{aligned}$$

iii) Fix $x \in \mathbb{R}$ and let $h_n \downarrow 0 \quad (h_n \geq 0 \quad h_n \to 0, h_1 \geq h_2 \geq \dots)$
Define $C_n = \{X \leq x + h_n\}$
then $C_1 \supseteq C_2 \supseteq \dots$

$$\lim_{n\to\infty} C_n = \bigcap_{n=1}^{\infty} C_n = \{X \leq x\}$$

Therefore

$$\lim_{n\to\infty} F(x+h_n) = \lim_{n\to\infty} P(C_n) = P(\bigcap_{n=1}^{\infty} C_n) = P(X \leq x) = F(x)$$

Hence

$$\lim_{h\downarrow 0} F(x+h) = F(x)$$

So $F$ is right-continuous.

□

**Def. 3.5 The Inverse CDF**

The inverse CDF of a r.v. $X$ is defined by

$$F^{-1}(y) = \sup\{x : F(x) < y\}$$

**Theorem 3.5.1** If $F$ satisfies properties i) to iii) of Thm 3.4, then such an $F$ is the CDF of some r.v.

**Theorem 3.6** If $F$ is a CDF, it can only have countably many discontinuities.

# Lecture 4: Random Variables & Distributions

Sep 17, 2026

**Def. 3.7 Density Function**

A random variable has a density function

$$f : \mathbb{R} \to \mathbb{R}, \quad \text{if for all } x \in \mathbb{R}$$

$$F(x) = \int_{-\infty}^{x} f(y)\, dy$$

we call $f(\cdot)$ the density function.

**Def. 3.8 Joint Distribution**

We define the joint distribution function of a random vector $\overrightarrow{X} = (X_1, \dots, X_n)$ by

$$F(x_1, \dots, x_n) = P(X_1 \leq x_1, \dots, X_n \leq x_n)$$

using the notation $P(A, B) \overset{def}{=} P(A \bigcap B)$

**Theorem 3.8.1** A joint distribution function $F : \mathbb{R}^n \to \mathbb{R}$ satisfies the following properties:

i) if $(x_1, \dots, x_n)$ and $(x_1', \dots, x_n')$ satisfy $x_i \leq x_i'$ for all $i \in [n]$,

$$F(x_1, \dots x_n) \leq F(x_1', \dots x_n')$$

ii) $\lim_{x_i \to -\infty} F(x_1, \dots, x_n) = 0$ for all $i \in \{1, \dots, n\}$ and $\lim_{x_1 \to \infty, \dots x_n \to \infty} F(x_1, \dots, x_n) = 1$

iii) $\lim_{h \to 0^+} F(x_1 + h, \dots, x_n) = \cdots = \lim_{h \to 0^+} F(x_1, \dots x_n + h)$

$$= F(x_1, \dots, x_n)$$

Proofs are analogous to the proofs under the univariate case.

**Def. 3.9 The Joint Density**

A random vector has a joint density function:

$$f : \mathbb{R}^n \to \mathbb{R}, \quad \text{if } \forall (x_1, \dots, x_n) \in \mathbb{R}^n$$

$$F(x_1, \dots x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f(y_1, \dots, y_n)\, dy_n \dots\, dy_1$$

**Theorem 3.10 Independence of Random Variables**

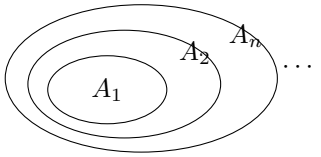
Random variables $X_1, \dots, X_n$ with CDFs $F_1, \dots, F_n$ are independent if and only if

$$F(x_1, \dots, x_n) = \prod_{i=1}^{n} F_i(x_i), \quad \forall (x_1, \dots, x_n) \in \mathbb{R}^n$$

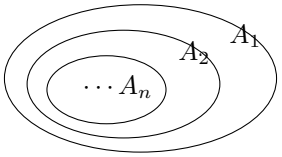
**Limits of sets:**

Let $A_i$ be sets for $i \in \{1, 2, \dots\}$ and consider $\lim_{n \to \infty} A_n$.

- If $A_1 \subseteq A_2 \subseteq \cdots \subseteq A_n \subseteq \ldots$ (increasing), then $\lim_{n\to\infty} \bigcup_{i=1}^{n} A_i = \bigcup_{i=1}^{\infty} A_i$.
- If $A_1 \supseteq A_2 \supseteq \cdots \supseteq A_n \supseteq \ldots$ (decreasing), then $\lim_{n\to\infty} \bigcap_{i=1}^{n} A_i = \bigcap_{i=1}^{\infty} A_i$.



increasing: $A_1 \subseteq A_2 \subseteq \cdots$



decreasing: $A_1 \supseteq A_2 \supseteq \cdots$

# Lecture 5: Expectation

Sep 22, 2026

## Expectation

In STA237:

$$E(X) = \begin{cases} \sum_x x \cdot p(x) & \text{if X discrete} \\ \int x f(x)\, dx & \text{if X continuous} \end{cases}$$

for this course, we are defining probability via "measure theory", so we need to revisit the definition of expectation.

Particularly, under the "measure theory" perspective:

$$E(X) = \int x\, dP \quad \Rightarrow P = \text{probability measure}$$

$$\int x f(x)\, dx \quad \Rightarrow \quad \int x\, dP$$

"Riemann Integral"

**Def. 5.1 Simple Expectation**

If $X$ only takes finitely many values $x_1, \ldots, x_n$, and $A_i = \{\omega : X(\omega) = x_i\}$, define the expectation of $X$ by:

$$E(X) = \sum_{i=1}^{n} x_i P(A_i)$$

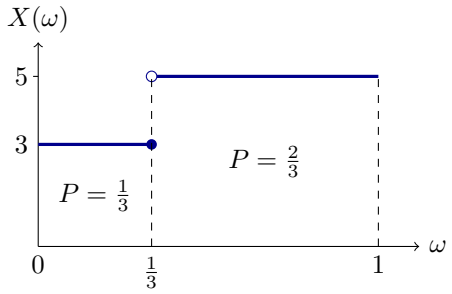
**Example 5.1.1** For an arbitrary set $A \subset \Omega$, let

$$Y(\omega) = I\{\omega \in A\} \qquad \text{Then } E(Y) = P(A)$$

$$\begin{aligned} E(Y) &= \sum_{i=0}^{1} y_i \cdot P(A_i) \qquad Y \text{ only takes two values 0 and 1} \\ &= 1 \cdot P\{\omega \in A\} + 0 \cdot P\{\omega \notin A\} \\ &= P(A) = P(\{\omega : \omega \in A\}) \end{aligned}$$

**Example 5.1.2** $P$ is uniform measure on $\Omega = [0, 1)$, and

$$X(\omega) = \begin{cases} 5 & \omega > \frac{1}{3} \\ 3 & \omega \leq \frac{1}{3} \end{cases}$$



$$E(X) = 5 \cdot P(\omega > \frac{1}{3}) + 3 \cdot P(\omega \leq \frac{1}{3})$$
$$= 5 \cdot (1 - \frac{1}{3}) + 3 \cdot \frac{1}{3}$$
$$= \frac{13}{3}$$

**Proposition 5.2 Properties of Expectation**
If $X$ and $Y$ are simple random variables, the following hold:

i) If $X \geq 0$ a.s., $E(X) \geq 0$
ii) For any $a \in \mathbb{R}$, $E(aX) = a \cdot E(X)$
iii) $E(X + Y) = E(X) + E(Y)$

Remark: "a.s." means almost surely. if $E$ a.s. $\Rightarrow P(E) = 1$
$X \geq 0$ a.s. $\Rightarrow P(X \geq 0) = 1 \Leftrightarrow P(\{\omega : X(\omega) \geq 0\}) = 1$
we may have certain $\omega$'s s.t. $X(\omega) < 0$, but $P(\{\omega : X(\omega) < 0\}) = 0$

*Proof.* Both i) and ii) are trivial.
For iii), let $X$ take values $x_1, \ldots, x_n$ on sets $A_1, \ldots, A_n$ (i.e. $A_i = \{\omega : X(\omega) = x_i\}$) and $Y$ takes values $y_1, \ldots, y_m$ on sets $B_1, \ldots, B_m$.
Observe that there are $m \cdot n$ events

$$C_{ij} = \left\{X = x_i \bigcap Y = y_j\right\}$$

Then

$$\begin{aligned} E(X + Y) &= \sum_{i=1}^{n} \sum_{j=1}^{m} (x_i + y_j) \cdot P(C_{ij}) \\ &= \sum_{i=1}^{n} x_i \sum_{j=1}^{m} P(X = x_i \cap Y = y_j) + \sum_{j=1}^{m} y_j \sum_{i=1}^{n} P(X = x_i \cap Y = y_j) \\ &= \sum_{i=1}^{n} x_i P(X = x_i) + \sum_{j=1}^{m} y_j P(Y = y_j) \\ &= E(X) + E(Y) \end{aligned}$$

□

**Lemma 5.2.1** If Proposition 5.2 holds, the following properties also hold:

iv) If $X \leq Y$ a.s., then $E(X) \leq E(Y)$
v) If $X = Y$ a.s., then $E(X) = E(Y)$
vi) $|E(X)| \leq E(|X|)$

proof as exercise.

Our logic is, we first define

expectation of simple random variables

$\Downarrow$

expectation of bounded R.V.'s (approximation)

$\Downarrow$

expectation of non-negative R.V's (truncation)

$\Downarrow$

expectation of any R.V.'s (symmetric)

**Def. 5.3 Bounded Random Variable**

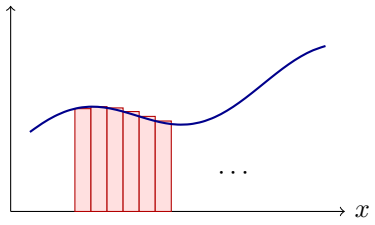
A random variable $X$ is bounded if there exists $M < \infty$, such that $|X| \leq M$ a.s.

**Def. 5.3.1 Expectation of Bounded Random Variable**

If $X$ is a bounded R.V., then

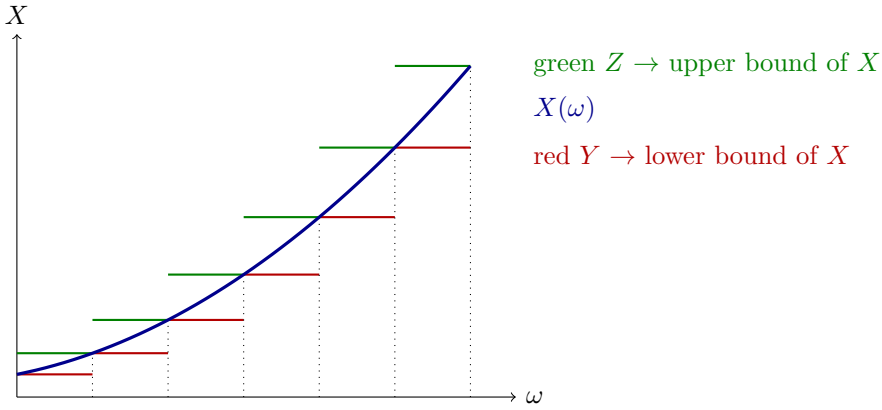
$$\begin{aligned} E(X) &= \inf \{E(Z) : Z \text{ simple}, Z \geq X \text{ a.s.}\} \\ &= \sup \{E(Y) : Y \text{ simple}, Y \leq X \text{ a.s.}\} \end{aligned}$$

Recall, when we defined the integral:
the integral is defined as the limit value of sums of areas of tiny rectangles.




When it comes to the Riemann integral, we are no longer cutting $X$ axis into tiny intervals, but cutting each set into tiny subsets and calculate the sum of $x$ times the measure of each tiny subset.

$$\begin{aligned} E(X) &= \inf \{E(Z) : \quad Z \text{ is simple and } Z \geq X \text{ a.s.}\} \\ &= \sup \{E(Y) : \quad Y \text{ is simple and } Y \leq X \text{ a.s.}\} \end{aligned}$$



"Squeeze the R.V. $X$ via two simple R.V.'s $Y$ and $Z$, with one capturing the upper bound and the other capturing the lower bound".

*Proof.* Since $X$ is bounded, $|X| < M$ a.s.

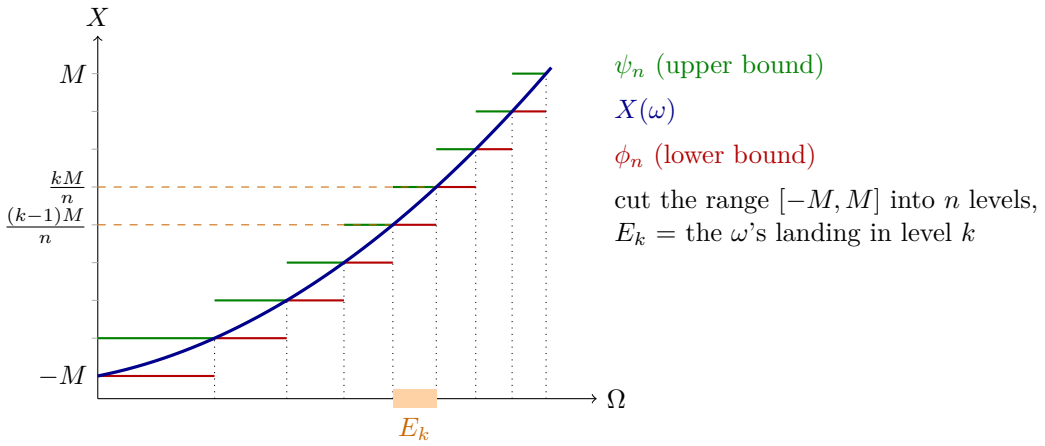
$$L = \sup\{EY : \quad Y \text{ simple and } Y \le X \text{ a.s.}\}$$

$$U = \inf\{EZ : \quad Z \text{ simple and } Z \ge X \text{ a.s.}\}$$

$$L \le U \text{ as } Y \le X \text{ a.s. and } Z \ge X \text{ a.s.}$$

Our goal is to prove $L \ge U \quad \Rightarrow \quad L = U \overset{def}{=} E(X)$
the way to cut $[-M, M]$ into tiny intervals.
for any arbitrary $n$, we define

$$E_k = \left\{\omega : \frac{kM}{n} \ge X(\omega) \ge \frac{(k-1)M}{n}\right\}$$




we then define two simple R.V.'s
Lower bound:

$$\phi_n(\omega) = \sum_k \frac{(k-1)M}{n} \cdot I(\omega \in E_k)$$

Upper bound:

$$\psi_n(\omega) = \sum_k \frac{kM}{n} \cdot I(\omega \in E_k)$$

$$\phi_n(\omega) \le X(\omega) \le \psi_n(\omega)$$

$$E\phi_n(\omega) \le E(X) \le E\psi_n(\omega)$$

$$E\psi_n(\omega) = E\phi_n(\omega) + \frac{M}{n}$$

Let $n \to \infty$, $\quad \frac{M}{n} \to 0$, so the upper bound and lower bound converge:

$$L \ge E\phi_n = E\psi_n - \frac{M}{n} \ge U - \frac{M}{n} \quad \Rightarrow \quad L \ge U$$

$$E(X) = E\psi_n(\omega) = E\phi_n(\omega) \quad \text{as } n \to \infty$$

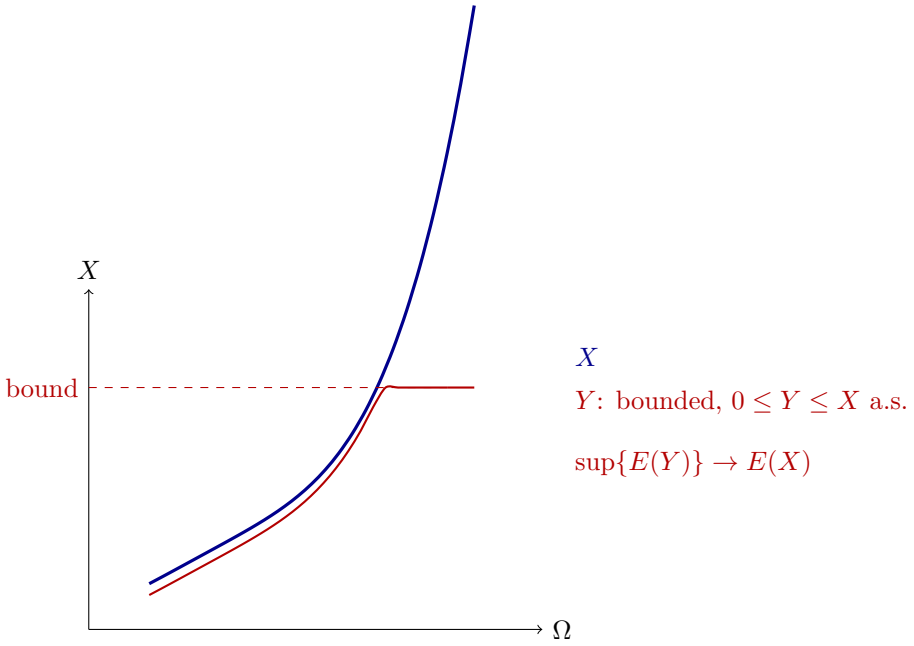
□

$X \leq Y$ a.s. means:

step 1. $\omega : X(\omega) \leq Y(\omega)$
step 2. $\{\omega : X(\omega) \leq Y(\omega)\}$
step 3. $P(\{\omega : X(\omega) \leq Y(\omega)\}) = 1$

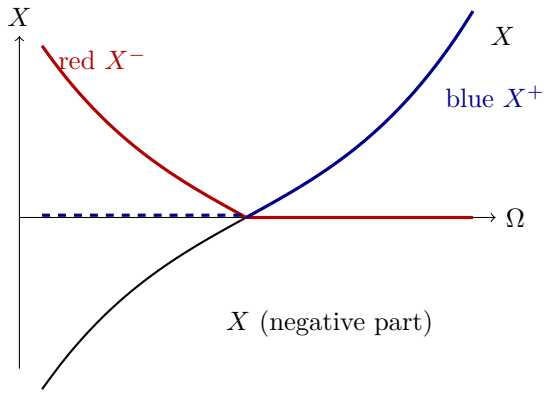
then we say $X \leq Y$ a.s.

**Def. 5.4 Non-negative Random Variable**
If $X \geq 0$ a.s. $(P(\{\omega : X(\omega) \geq 0\}) = 1)$
define $E(X) = \sup\{E(Y) : Y \text{ bounded}, 0 \leq Y \leq X \text{ a.s.}\}$




Remark: Def 5.4. only provides a definition of $E(X)$ when $X$ is non-negative, it does not guarantee $E(X) < \infty$.

**Def. 5.5 Integrable Random Variables**
For any random variable $X$, let $X^+(\omega) = \max\{X(\omega), 0\}$, and $X^-(\omega) = \max\{-X(\omega), 0\}$.
Observe that both $X^+$ and $X^-$ are non-negative and $X = X^+ - X^-$.
Then, if $E(|X|) = E(X^+) + E(X^-) < \infty$, we say R.V. $X$ is integrable and define $E(X) = E(X^+) - E(X^-)$




1. $E|X| = E(X^+) + E(X^-) < \infty \Rightarrow X$ is integrable
2. $E(X) = E(X^+) - E(X^-)$

**Proposition 5.6 Properties of Expected Values**
If R.V. $X$ is either bounded, non-negative or integrable, Proposition 5.2 holds.

1. $X$ and $Y$ may not need to be simple, if they are bounded, non-negative or integrable, the 3 properties (prop 5.2) also hold.
2. Lemma 5.2.1 says, if prop 5.2 holds, the additional 3 properties also hold.
3. To sum up, if R.V.'s are bounded, non-negative or integrable, all the 6 properties hold.

Hint for Assignment 1. Q2:

$E_1$ and $E_2$ are independent.
$E_1^c$ and $E_2^c$ are independent.

if $E_1, E_2, \dots E_n$, you want to prove $E_1^c, E_2^c, \dots E_n^c$ are independent.

$$P(E_1^c \bigcap E_2^c \bigcap \cdots \bigcap E_n^c) = \cdots$$

you can flip 1 event at a time

first try to flip the first...

$$P(E_1^c \bigcap E_2 \bigcap E_3 \bigcap \cdots \bigcap E_n)$$

$$= P(E_2 \bigcap E_3 \bigcap \cdots \bigcap E_n) - P(E_1 \bigcap E_2 \bigcap \cdots \bigcap E_n)$$

$$= \cdots = (1 - P(E_1))P(E_2) \cdots P(E_n)$$

Then you flip the second one...

$$P(E_1^c \bigcap E_2^c \bigcap E_3 \bigcap \cdots \bigcap E_n)$$

Via induction, complete the proofs

# Lecture 6: Independence and Expectation, Variance

Sep 24, 2026

**Theorem 5.7 Independence and Expectation**
If $X$ and $Y$ are independent with $EX, EY < \infty$, then

$$E(XY) = (EX)(EY)$$

*Proof.* First, consider both $X$ and $Y$ are simple, so that

$$X = \sum_{i=1}^{n} x_i I(A_i) \quad \text{and} \quad Y = \sum_{j=1}^{m} y_j I(B_j)$$

$$\begin{aligned}
E(XY) &= E\Big[\sum_{i=1}^{n}\sum_{j=1}^{m} x_i y_j \cdot I\{A_i \cap B_j\}\Big] \\
&= \sum_{i=1}^{n}\sum_{j=1}^{m} x_i y_j \underbrace{P(X \in A_i, Y \in B_j)}_{= P(X \in A_i)\cdot P(Y \in B_j) \text{ because } X \text{ and } Y \text{ indep.}} \\
&= \sum_{i=1}^{n}\sum_{j=1}^{m} x_i y_j P(\{\omega : X(\omega) \in A_i, Y(\omega) \in B_j\}) \\
&= \sum_{i=1}^{n}\sum_{j=1}^{m} x_i y_j P(X \in A_i) \cdot P(Y \in B_j) \\
&= \underbrace{\sum_{i=1}^{n} x_i \cdot P(X \in A_i)}_{EX} \underbrace{\sum_{j=1}^{m} y_j \cdot P(Y \in B_j)}_{EY} \\
&= (EX)(EY)
\end{aligned}$$

Next, consider the case that $X$ and $Y$ are bounded and non-negative, then we let $Z_1 \geq X, Z_2 \geq Y$, where $Z_1$ and $Z_2$ are simple. Notice that

$$Z_1 \geq X, Z_2 \geq Y \text{ implies } Z_1 \cdot Z_2 \geq XY$$

$$\begin{aligned}
(EX)(EY) &= \inf\{EZ_1 : Z_1 \text{ simple, } Z_1 \geq X \text{ a.s.}\} \\
&\quad \cdot \inf\{EZ_2 : Z_2 \text{ simple, } Z_2 \geq Y \text{ a.s.}\} \\
&= \inf\{(EZ_1)\cdot(EZ_2)\} \qquad \text{because } Z_1, Z_2 \text{ simple, so } \inf(EZ_1)\cdot\inf(EZ_2) = \inf(EZ_1 \cdot EZ_2) \\
&= \inf\{EZ_1Z_2 : Z_1 \text{ and } Z_2 \text{ simple and independent, } Z_1Z_2 \geq XY \text{ a.s.}\}
\end{aligned}$$

Additionally, if $Z_1$ and $Z_2$ are simple and

$$Z_1 \leq X \text{ a.s.} \qquad Z_2 \leq Y \text{ a.s.}$$

Similarly,

$$(EX)(EY) = \sup\{EZ_1Z_2 : Z_1 \text{ and } Z_2 \text{ are simple and independent, } Z_1 \le X, Z_2 \le Y \text{ a.s.}\}$$

Previously (Def 5.3.1), we proved that

$$\begin{aligned}&\inf\{EZ_1Z_2 : \quad Z_1, Z_2 \text{ are simple and independent with } Z_1 \ge X, Z_2 \ge Y \text{ a.s. }\}\\ =&\sup\{EZ_1Z_2 : \quad Z_1, Z_2 \text{ are simple and independent with } Z_1 \le X, Z_2 \le Y \text{ a.s. }\} = E(XY)\end{aligned}$$

$$\Rightarrow (EX)(EY) = E(XY)$$

Finally, for $X$ and $Y$ that are integrable, first realize that if $X$ and $Y$ are independent, then $X^+, X^-, Y^+, Y^-$ are also independent.

$$X = X^+ - X^- \qquad Y = Y^+ - Y^-$$

$$\begin{aligned}E(XY) &= E[(X^+ - X^-)(Y^+ - Y^-)]\\ &= E[X^+Y^+ - X^+Y^- - X^-Y^+ + X^-Y^-]\\ &= E(X^+Y^+) - E(X^+Y^-) - E(X^-Y^+) + E(X^-Y^-)\\ &= E(X^+)E(Y^+) - E(X^+)E(Y^-) - E(X^-)E(Y^+) + E(X^-)E(Y^-)\\ &= (EX)(EY)\end{aligned}$$

□

**Def. 5.8 Variance and Covariance**

The variance of a random variable $X$ is defined as

$$\mathrm{Var}(X) = E[(X - EX)^2]$$

The covariance of random variables $X$ and $Y$ is

$$\mathrm{Cov}(X, Y) = E[(X - EX)(Y - EY)]$$

The correlation coefficient is

$$\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}$$

# Lecture 7: Chapter 4 - Inequalities

Sep 29, 2026

**Theorem 4.1 Markov's Inequality**
If $X \geq 0$ a.s., then for all $a > 0$,

$$P(X \geq a) \leq \frac{E(X)}{a}$$

*Proof.* Define $Z(\omega) = a \cdot I\{X(\omega) \geq a\}$. Then, $Z \leq X$ a.s., and since $Z$ is simple,

$$E(Z) = a \cdot P(X \geq a)$$

$$P(X \geq a) = \frac{E(Z)}{a} \leq \frac{E(X)}{a}$$

□ □

**Theorem 4.2 Chebyshev's Inequality**
For all $t > 0$,

$$P(|X - E(X)| \geq t) \leq \frac{\text{Var}(X)}{t^2}$$

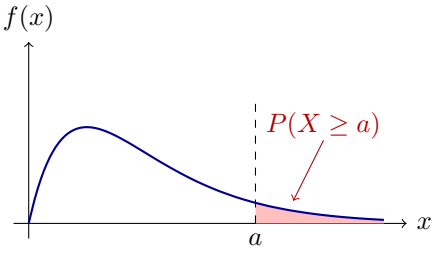
*Proof.* Apply Markov's inequality to $Y = [X - E(X)]^2 \geq 0$:

$$P(|X - E(X)| \geq t) = P([X - E(X)]^2 \geq t^2) \leq \frac{E[(X - E(X))^2]}{t^2} = \frac{\text{Var}(X)}{t^2}$$

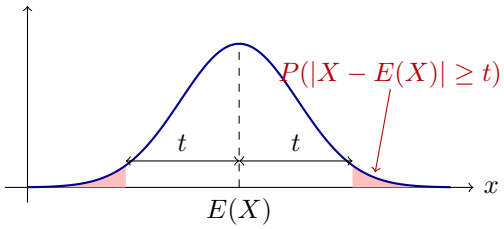
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Both Chebyshev and Markov's inequality talk about

$$P(X \geq \cdots) \quad \Leftarrow \quad \text{the tail probability}$$




Markov: uses the 1st moment




Chebyshev: uses the 2nd moment

"Concentration of Probability Mass"

Where Markov uses the 1st moment, Chebyshev uses 2nd moment.
What if we use other transformation/moments to quantify the tail probability?
E.g. the exponential moment? ⇒ Chernoff's Inequality

$$M_X(t) = E[e^{tX}]$$

**Theorem 4.3 Chernoff's Inequality**
For all $t \geq 0$, $\quad P(X \geq E(X) + t) \leq \inf_{\lambda > 0} M_{X-E(X)}(\lambda) e^{-\lambda t}$

$$P(X \geq E(X) + t) \text{ roughly } P(\underbrace{X - E(X)}_{\substack{\text{distance between } X \\ \text{to its expected value}}} \geq t)$$

$$\leq \inf_{\lambda > 0} M_{X-E(X)}(\lambda) e^{-\lambda t}$$

*Proof.* the function $g(x) = e^x$ is monotonically increasing and non-negative.

$$P(X - E(X) \geq t)$$

$$= P(e^{X-E(X)} \geq e^t)$$

$$= P(e^{\lambda(X-E(X))} \geq e^{\lambda t}) \quad \lambda > 0$$

Since $e^{\lambda(X-E(X))} \geq 0$ a.s., we can apply Markov's inequality

$$\leq \frac{E[e^{\lambda(X-E(X))}]}{e^{\lambda t}} \quad \text{where the numerator is the mgf } M_{X-E(X)}(\lambda)$$

We proved that

$$P(X - E(X) \geq t) \leq e^{-\lambda t} \cdot M_{X-E(X)}(\lambda)$$

Since this inequality holds for any arbitrary $\lambda > 0$, we can take the inf:

$$P(X - E(X) \geq t) \leq \inf_{\lambda > 0} e^{-\lambda t} M_{X-E(X)}(\lambda)$$

□ □

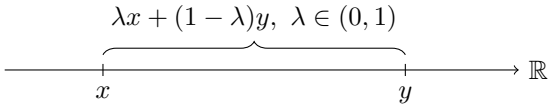
## Hoeffding's Inequality (Thm 4.4)

**Def. 4.4.1 Convex Function**
A function $f : \mathbb{R} \to \mathbb{R}$ is convex if for all $\lambda \in (0, 1)$ and $x, y \in \mathbb{R}$

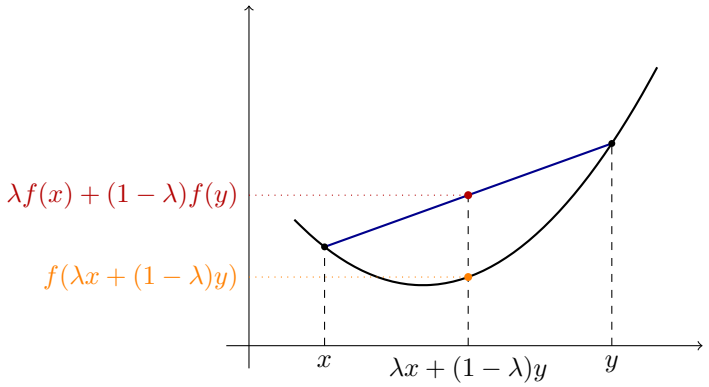
$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda) f(y)$$

Any value between $x$ and $y$ can be represented as

$$\lambda x + (1 - \lambda)y \text{ for } \lambda \in (0, 1)$$




if $\lambda \to 1$, then the value is closer to $x$. Vice versa for $\lambda \to 0$, the value is closer to $y$.




Convexity: $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y)$
$\Rightarrow$ the function is under the straight line

**Lemma 4.4.2 Hoeffding's Lemma**
If $Y$ satisfies $E(Y) = 0$ and $a \leq Y \leq b$ a.s., then for all $\lambda > 0$

$$M_Y(\lambda) \leq e^{\lambda^2 (b-a)^2/8}$$

*Proof.* Since $a \leq Y \leq b$ a.s., $Y = \alpha \cdot a + (1-\alpha) \cdot b$.
Transforming $Y \Rightarrow \alpha$, where $0 \leq \alpha \leq 1$,

$$\alpha = \frac{b-Y}{b-a}, \qquad 1-\alpha = \frac{Y-a}{b-a}$$

Also, since the function $e^{\lambda y}$ is convex,

$$e^{\lambda(\alpha a + (1-\alpha)b)} \leq \alpha e^{\lambda a} + (1-\alpha) e^{\lambda b}$$

Apply $f(y) = e^{\lambda y}$ transformation to r.v. $Y$:

$$e^{\lambda Y} \leq \alpha e^{\lambda a} + (1-\alpha) e^{\lambda b} = \frac{b-Y}{b-a} e^{\lambda a} + \frac{Y-a}{b-a} e^{\lambda b}$$

Take expectation on both sides:

$$E[e^{\lambda Y}] \leq E\Big[\frac{b - \overbrace{Y}^{\text{random var.}}}{b-a} \underbrace{e^{\lambda a}}_{\text{constant}}\Big] + E\Big[\frac{Y-a}{b-a} \underbrace{e^{\lambda b}}_{\text{constant}}\Big]$$

$$= \frac{b - E(Y)}{b-a} e^{\lambda a} + \frac{E(Y) - a}{b-a} e^{\lambda b}$$

$$= \underbrace{\frac{b}{b-a}}_{p} e^{\lambda a} + \frac{-a}{b-a} e^{\lambda b} \qquad (\text{since } E(Y) = 0)$$

Next, we let $p = \frac{b}{b-a}$:

$$E[e^{\lambda Y}] \leq p e^{\lambda a} + (1-p) e^{\lambda b}$$

We take log on RHS:

$$\log\{E[e^{\lambda Y}]\} \le \log\{pe^{\lambda a} + (1-p)e^{\lambda b}\}$$
$$= \log\{e^{\lambda a}[p + (1-p)e^{\lambda(b-a)}]\}$$
$$= \underbrace{\lambda a}_{=(p-1)u} + \log[p + (1-p)e^{\lambda(b-a)}]$$
$$(p-1)u = \left(\frac{b}{b-a} - 1\right)(b-a)\lambda = \frac{a}{b-a}(b-a)\lambda = a\lambda$$

Next, let $u = (b-a)\lambda$:

$$\log\{E[e^{\lambda Y}]\} \le (p-1)u + \log[p + (1-p)e^u] =: \varphi(u)$$
$$\varphi(0) = 0, \qquad \varphi'(u) = (p-1) + \frac{(1-p)e^u}{p + (1-p)e^u}, \qquad \varphi'(0) = 0$$
$$\varphi(u) = \underbrace{\varphi(0)}_{0} + \underbrace{\varphi'(0)}_{0}u + \frac{1}{2}\varphi''(\xi)u^2 \qquad \text{(Taylor's Expansion at } u = 0)$$

For some $\xi$ between 0 and $u$, $\quad \varphi(u) = \frac{1}{2}\varphi''(\xi)u^2$

$$\varphi''(u) = \frac{p(1-p)e^u}{[p + (1-p)e^u]^2} \le \frac{1}{4}$$
$$\Rightarrow \varphi(u) \le \frac{1}{8}u^2$$
$$\Rightarrow \log\{E[e^{\lambda Y}]\} \le \frac{1}{8}u^2 = \frac{(b-a)^2\lambda^2}{8} \quad \text{as } u = (b-a)\lambda$$
$$\Rightarrow M_Y(\lambda) \le e^{\frac{(b-a)^2\lambda^2}{8}}$$

□ □

**Theorem 4.4 Hoeffding's Inequality**
Suppose $X_1, X_2, \ldots, X_n, \ldots$ are indep. with $a_i \le X_i \le b_i$ for all $i$.
Then letting $S_n = \sum_{i=1}^n X_i$, for all $t > 0$,

$$P(|S_n - E(S_n)| \ge t) \le 2 \cdot \exp\left\{-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right\}$$

*Proof.* We first prove the $P(S_n - E(S_n) \ge t)$ part.
Applying Chernoff's inequality,

$$P(\underbrace{S_n - E(S_n)}_{(1)} \ge t) \le e^{-\lambda t}\underbrace{E[e^{\lambda(S_n - E(S_n))}]}_{(2)}$$

(1): $\quad S_n - E(S_n) = \sum_{i=1}^n (X_i - E(X_i)) \quad$ as $S_n = \sum_{i=1}^n X_i$
(2):

$$e^{\lambda(S_n - E(S_n))} = e^{\lambda\sum_{i=1}^n (X_i - E(X_i))} = \prod_{i=1}^n e^{\lambda[X_i - E(X_i)]}$$

Taking expectations, the factors are independent so $E$ of the product = product of $E$'s:

$$M_{S_n - E(S_n)}(\lambda) = \prod_{i=1}^{n} M_{X_i - E(X_i)}(\lambda)$$

Applying Hoeffding's Lemma to $Y = X_i - E(X_i)$: $\quad M_{X_i - E(X_i)}(\lambda) \leq e^{\frac{(b_i - a_i)^2 \lambda^2}{8}}$

$$M_{S_n - E(S_n)}(\lambda) \leq \prod_{i=1}^{n} e^{\frac{(b_i - a_i)^2 \lambda^2}{8}} = e^{\frac{\lambda^2}{8} \sum_{i=1}^{n} (b_i - a_i)^2}$$

Combine with Chernoff's inequality:

$$P(S_n - E(S_n) \geq t) \leq e^{-\lambda t + \frac{\lambda^2}{8} \sum_{i=1}^{n} (b_i - a_i)^2}$$

Rearrange, we have Hoeffding's inequality. □ □

# Lecture 8:

Oct 1, 2026

**Theorem 4.5 Jensen's Inequality**

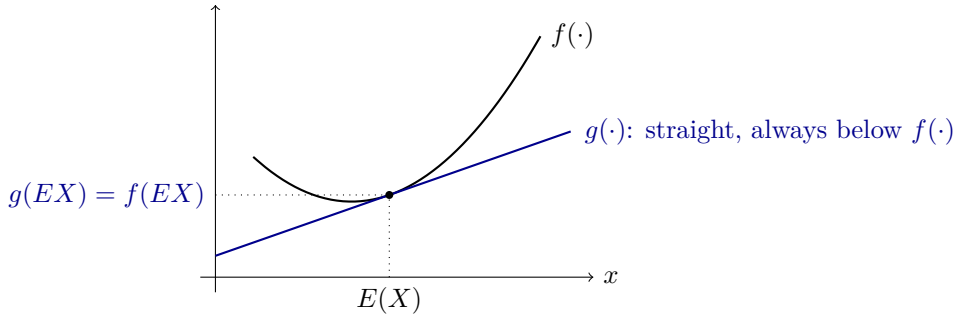
If $f$ is a convex function and $X$ is a random variable such that $f(X)$ is integrable,

$$f(EX) \leq Ef(X)$$

The inequality is **flipped** if $f$ is **concave**.

*Proof.* Since $f$ is convex, there exists a function $g$ s.t.

$$\begin{aligned} g(x) &= ax + b \text{ and} && \Leftarrow \quad g(\cdot) \text{ is a straight line} \\ g(x) &\leq f(x) \text{ for all } x \text{ and} && \Leftarrow \quad g(\cdot) \text{ always below } f(\cdot) \\ g(EX) &= f(EX) && \Leftarrow \quad g(\cdot) = f(\cdot) \text{ at } x = EX \end{aligned}$$




$$f(x) \geq g(x) \text{ for all } x$$

$$\begin{aligned} E[f(X)] \geq E[g(X)] &= E[aX + b] \\ &= aE(X) + b \\ &= g(E(X)) \\ &= f(E(X)) \end{aligned}$$

□

**Example 4.5.1 Applying the Jensen's Inequality**

$$[E(X)]^2 \leq E[X^2] \quad f(x) = x^2 \Rightarrow \text{convex}$$

$$E[\log X] \leq \log[EX] \quad f(x) = \log x \Rightarrow \text{concave}$$

**Lemma 4.6 Young's Inequality**

For any $p, q > 1$ and $\frac{1}{p} + \frac{1}{q} = 1$,

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q} \quad \text{for any } x, y \geq 0$$

*Proof.* Let $h_y(x) = \frac{1}{p}x^p + \frac{1}{q}y^q - xy$
To prove $h_y(x) \geq 0$, we take derivatives

$$h'_y(x) = \frac{dh_y(x)}{dx} = x^{p-1} - y$$

$$h''_y(x) = \frac{d^2h_y(x)}{dx^2} = (p-1)x^{p-2} \geq 0, \text{ because } x \geq 0$$

Therefore $h'_y(x) = 0 \Rightarrow x^* = y^{\frac{1}{p-1}}$ is the minimizer.
Plugging in,

$$\begin{aligned} h_y(x^*) &= \frac{1}{p}\left(y^{\frac{1}{p-1}}\right)^p + \frac{1}{q}y^q - y^{\frac{1}{p-1}} \cdot y \\ &= \frac{1}{p}y^{\frac{p}{p-1}} + \frac{1}{q}y^q - y^{\frac{p}{p-1}} \end{aligned}$$

Since

$$\frac{1}{p} + \frac{1}{q} = 1 \;\Rightarrow\; \frac{1}{q} = 1 - \frac{1}{p} \;\Rightarrow\; q = \frac{p}{p-1}$$

substituting $q = \frac{p}{p-1}$ and $\frac{1}{q} = 1 - \frac{1}{p}$:

$$\begin{aligned} h_y(x^*) &= \frac{1}{p}y^{\frac{p}{p-1}} + \left(1 - \frac{1}{p}\right)y^{\frac{p}{p-1}} - y^{\frac{p}{p-1}} \\ &= 0 \end{aligned}$$

$$\Rightarrow h_y(x) \geq h_y(x^*) = 0$$

Therefore $\frac{1}{p}x^p + \frac{1}{q}y^q \geq xy$.
(If $p = q = 2$: $\frac{1}{2}x^2 + \frac{1}{2}y^2 \geq xy \;\Leftrightarrow\; x^2 + y^2 \geq 2xy$.) □

**Theorem 4.6 Hölder's Inequality**
If $p, q > 1$ s.t. $\frac{1}{p} + \frac{1}{q} = 1$,

$$E[|XY|] \leq [E(|X|^p)]^{\frac{1}{p}} \cdot [E(|Y|^q)]^{\frac{1}{q}}$$

*Proof.* Let

$$X^* = \frac{|X|}{[E(|X|^p)]^{\frac{1}{p}}} \quad \text{and} \quad Y^* = \frac{|Y|}{[E(|Y|^q)]^{\frac{1}{q}}}$$

Observe that

$$E[(X^*)^p] = E[(Y^*)^q] = 1$$

since

$$E[(X^*)^p] = E\left[\left(\frac{|X|}{[E(|X|^p)]^{\frac{1}{p}}}\right)^p\right] = E\left[\frac{|X|^p}{E(|X|^p)}\right] = \frac{E(|X|^p)}{E(|X|^p)} = 1$$

and the same for $Y^*$ with $q$.

Applying the Young's Inequality

$$X^*Y^* \leq \frac{(X^*)^p}{p} + \frac{(Y^*)^q}{q}$$

Take expectations on both sides

$$E[X^*Y^*] \leq \frac{E[(X^*)^p]}{p} + \frac{E[(Y^*)^q]}{q}$$

$$E\left[\frac{|X||Y|}{E(|X|^p)^{\frac{1}{p}}E(|Y|^q)^{\frac{1}{q}}}\right] \leq \frac{1}{p} + \frac{1}{q} = 1$$

Therefore, re-arrange

$$E[|X||Y|] \leq E(|X|^p)^{\frac{1}{p}}E(|Y|^q)^{\frac{1}{q}}$$

□

**Proposition 4.6.1 Cauchy-Schwarz Inequality**

$$E[|XY|] \leq \sqrt{E(X^2)E(Y^2)}$$

*Proof.* Exercise. □

**Theorem 4.7 Minkowski's Inequality**
If $p > 1$, then

$$[E(|X+Y|^p)]^{\frac{1}{p}} \leq [E(|X|^p)]^{\frac{1}{p}} + [E(|Y|^p)]^{\frac{1}{p}}$$

(The triangle inequality in probability)

**EXAM UPDATE: CAN BRING CHEATSHEET DOUBLE SIDE HANDWRITTEN!!!!**

# Lecture 9: Chapter 5 - Limiting Theorems and Convergences

Oct 6, 2026

**Theorem 5.1 Continuity of Measure**
If $\{A_n\} \uparrow A$ or $\{A_n\} \downarrow A$, then $\lim_{n\to\infty} P(A_n) = P(A)$.
Note:

1. $\{A_n\} \uparrow A$ means:

$$A_1 \subseteq A_2 \subseteq \cdots \subseteq A_n \subseteq \cdots \to A, \qquad \lim_{n\to\infty} \bigcup_{i=1}^{n} A_i = A$$

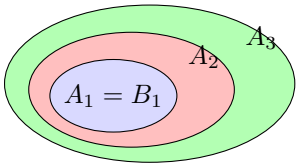
2. $\{A_n\} \downarrow A$ means:

$$A_1 \supseteq A_2 \supseteq \cdots \supseteq A_n \supseteq \cdots \to A, \qquad \lim_{n\to\infty} \bigcap_{i=1}^{n} A_i = A$$

Note: $A_1 \subseteq \cdots \subseteq A_n \subseteq \cdots \to A$ means

$$P(A_1) \leq \cdots \leq P(A_n) \leq \cdots \to P(A)$$

*Proof.* Suppose $\{A_n\} \uparrow A$.
We let $A_0 = \varnothing$, and $B_1 = A_1$, $B_n = A_n \cap A_{n-1}^c$.




red: $B_2 = A_2$ excluding $A_1$
green: $B_3 = A_3$ excluding $A_2$

Observe that:

1. $\bigcup_{i=1}^{n} A_i = \bigcup_{i=1}^{n} B_i$
2. $B_i$'s are disjoint.

Therefore, $P\left(\bigcup_{i=1}^{n} A_i\right) = P\left(\bigcup_{i=1}^{n} B_i\right)$, and

$$\begin{aligned} P(A) = P\left(\lim_{n\to\infty} \bigcup_{i=1}^{n} A_i\right) &= P\left(\lim_{n\to\infty} \bigcup_{i=1}^{n} B_i\right) = P\left(\bigcup_{i=1}^{\infty} B_i\right) \\ &= \lim_{n\to\infty} \sum_{i=1}^{n} P(B_i) \\ &= \lim_{n\to\infty} P(A_n) \end{aligned}$$

For the case $A_n \downarrow A$, the proof is similar; one may take the complement:

$$\{A_n\} \downarrow A \;\Rightarrow\; \{A_n^c\} \uparrow A^c$$

□

**Example 5.1.1** If each $A_n = \left[0, 1 - \frac{1}{n}\right]$, then

$$\lim_{n\to\infty} A_n = [0, 1) = A, \qquad A_n \uparrow A$$

If we apply a uniform measure $P(\cdot)$ on $\{A_n\}$,

$$P(A) = \lim_{n\to\infty} P(A_n) = \lim_{n\to\infty} \left(1 - \frac{1}{n}\right) = 1$$

**Def. 5.2 Tail Events**

Consider a sequence $A_1, A_2, \cdots \subseteq \Omega$, define the tail events as:

$$\limsup_{n\to\infty} A_n = \{A_n \text{ i.o.}\} = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k$$

$$\liminf_{n\to\infty} A_n = \{A_n \text{ a.a.}\} = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k$$

Note: the notations
i.o. means **infinitely often**
a.a. means **almost always**

$$\limsup_{n\to\infty} A_n = \{A_n \text{ i.o.}\} = \bigcap_{n=1}^{\infty} \underbrace{\bigcup_{k=n}^{\infty} A_k}_{\text{tail union}}$$

You want to find the smallest set that contains the union of your tail events:
"elements that will appear infinitely many times".

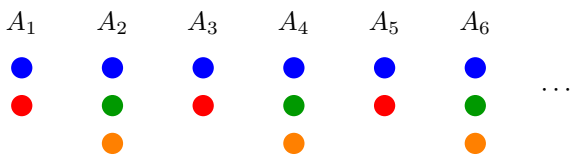
$$\liminf_{n\to\infty} A_n = \{A_n \text{ a.a.}\} = \bigcup_{n=1}^{\infty} \underbrace{\bigcap_{k=n}^{\infty} A_k}_{\text{tail intersect}}$$

You want to find the largest set that contains the intersect of your tail events:
"elements that will appear in all tail sets".

**Example Tail Events**

Suppose $\{A_n\}$ is constructed as follows:

$$A_i = \begin{cases} \{\text{blue}, \text{red}\} & i \text{ odd} \\ \{\text{blue}, \text{green}, \text{orange}\} & i \text{ even} \end{cases}$$



$$\limsup_{n\to\infty} A_n = \{A_n \text{ i.o.}\} = \{\text{blue}, \text{red}, \text{green}, \text{orange}\}$$

$$\liminf_{n\to\infty} A_n = \{A_n \text{ a.a.}\} = \{\text{blue}\}$$

$\{A_n \text{ i.o.}\} \Rightarrow$ "the elements that will appear infinitely often in the tail of $\{A_n\}$"
$\{A_n \text{ a.a.}\} \Rightarrow$ "the elements that will appear almost always in the tail of $\{A_n\}$"

**Corollary 5.2.1**

$$P(A_n \text{ i.o.}) = 1 - P(A_n^c \text{ a.a.})$$

**Proposition 5.2.2**

$$P(A_n \text{ a.a.}) \le \liminf_{n\to\infty} P(A_n) \le \limsup_{n\to\infty} P(A_n) \le P(A_n \text{ i.o.})$$

i.e.

$$P\left(\liminf_{n\to\infty} A_n\right) \le \liminf_{n\to\infty} P(A_n) \le \limsup_{n\to\infty} P(A_n) \le P\left(\limsup_{n\to\infty} A_n\right)$$

*Proof.* We prove the 1st "$\le$".

$$\liminf_{n\to\infty} A_n = \bigcup_{n=1}^{\infty} \underbrace{\bigcap_{k=n}^{\infty} A_k}_{B_n}$$

We let $B_n = \bigcap_{k=n}^{\infty} A_k$ ("the intersect of tail events"), so $\liminf_{n\to\infty} A_n = \bigcup_{n=1}^{\infty} B_n$.
Notice that $B_1 \subseteq B_2 \subseteq \cdots \subseteq B_n \subseteq \cdots$ as $B_n = \bigcap_{k=n}^{\infty} A_k$:

$$\begin{aligned} B_1 &= A_1 \cap A_2 \cap \cdots \cap A_n \cap \cdots \\ B_2 &= \quad A_2 \cap \cdots \cap A_n \cap \cdots \qquad \Rightarrow B_1 \subseteq B_2 \end{aligned}$$

So $B_n \uparrow B = \bigcup_{n=1}^{\infty} B_n$, and $\lim_{n\to\infty} P(B_n) = P(B)$ by continuity of measure (Thm 5.1).

$$\begin{aligned} P\left(\liminf_{n\to\infty} A_n\right) &= P\left(\bigcup_{n=1}^{\infty} B_n\right) = P\left(\lim_{n\to\infty} B_n\right) \\ &= \lim_{n\to\infty} P(B_n) && \text{as } B_n \text{ converges} \\ &= \liminf_{n\to\infty} P(B_n) = \liminf_{n\to\infty} P\left(\bigcap_{k=n}^{\infty} A_k\right) \\ &\le \liminf_{n\to\infty} P(A_n) && \text{because } \textstyle\bigcap_{k=n}^{\infty} A_k \subseteq A_n \end{aligned}$$

The 2nd "$\le$" is by definition.
The 3rd "$\le$" can be proved similarly. □

**Theorem 5.3 Borel-Cantelli Lemma** i) If $\sum_{n=1}^{\infty} P(A_n) < \infty$, then $P(A_n \text{ i.o.}) = 0$.
ii) If $\sum_{n=1}^{\infty} P(A_n) = \infty$ and $\{A_n\}$ are independent, then $P(A_n \text{ i.o.}) = 1$.

*Proof.* i)

$$\{A_n \text{ i.o.}\} = \bigcap_{n=1}^{\infty} \underbrace{\bigcup_{k=n}^{\infty} A_k}_{B_n}, \qquad B_n = \bigcup_{k=n}^{\infty} A_k, \qquad B_n \downarrow B$$

$$
\begin{aligned}
P(A_n \text{ i.o.}) &= P\left(\bigcap_{n=1}^{\infty}\bigcup_{k=n}^{\infty} A_k\right) = P\left(\bigcap_{n=1}^{\infty} B_n\right) \\
&= \lim_{n\to\infty} P(B_n) && \text{as } B_n \downarrow B = \textstyle\bigcap_{n=1}^{\infty} B_n \\
&= \lim_{n\to\infty} P\left(\bigcup_{k=n}^{\infty} A_k\right) \\
&\le \lim_{n\to\infty} \sum_{k=n}^{\infty} P(A_k) \\
&= 0 && \text{as } \textstyle\sum_{k=1}^{\infty} P(A_k) < \infty
\end{aligned}
$$

ii)

$$
\begin{aligned}
1 - P(A_n \text{ i.o.}) &= P\left(\{A_n \text{ i.o.}\}^c\right) \\
&= P\left(\bigcup_{n=1}^{\infty} \underbrace{\bigcap_{k=n}^{\infty} A_k^c}_{C_n}\right) && \text{because } \{A_n \text{ i.o.}\}^c = \textstyle\bigcup_{n=1}^{\infty}\bigcap_{k=n}^{\infty} A_k^c
\end{aligned}
$$

Let $C_n = \bigcap_{k=n}^{\infty} A_k^c$, $C_n \uparrow C$.

$$
\begin{aligned}
&= P\left(\bigcup_{n=1}^{\infty} C_n\right) = P\left(\lim_{n\to\infty} C_n\right) && \text{as } C_n \uparrow C \\
&= \lim_{n\to\infty} P(C_n) = \lim_{n\to\infty} P\left(\bigcap_{k=n}^{\infty} A_k^c\right) \\
&= \lim_{n\to\infty} \prod_{k=n}^{\infty} [1 - P(A_k)] && \text{because } \{A_n\} \text{ independent}
\end{aligned}
$$

Each factor $1 - P(A_k) \le 1$, but that alone doesn't give 0. The missing step: use $1 - x \le e^{-x}$:

$$
\prod_{k=n}^{\infty} [1 - P(A_k)] \le \prod_{k=n}^{\infty} e^{-P(A_k)} = \exp\left\{-\sum_{k=n}^{\infty} P(A_k)\right\} = e^{-\infty} = 0
$$

since $\sum_{k=n}^{\infty} P(A_k) = \infty$ for every $n$. So

$$
1 - P(A_n \text{ i.o.}) = 0 \;\Rightarrow\; P(A_n \text{ i.o.}) = 1
$$

□

**Example 5.3.1 Converse does NOT hold for i)**
Uniform measure and $A_n = \left[0, \frac{1}{n}\right]$. Then $A_n \downarrow A = \{0\}$, and

$$
\begin{aligned}
\{A_n \text{ i.o.}\} &= \bigcap_{n=1}^{\infty}\bigcup_{k=n}^{\infty} A_k = \bigcap_{n=1}^{\infty}\bigcup_{k=n}^{\infty}\left[0, \frac{1}{k}\right] \\
&= \{0\} = \lim_{n\to\infty} A_n
\end{aligned}
$$

$$P(\{A_n \text{ i.o.}\}) = P(\{0\}) = 0$$

but

$$\sum_{n=1}^{\infty} P(A_n) = \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

**Example 5.3.2 Independence is needed for ii)**

Define $C_1, C_2, \ldots$ to be results of a fair coin toss, such that $C_i = 1$ if the $i$-th toss resulted in H. Let $A_1, A_2, \ldots$ be events such that

$$A_j = \Big\{\text{the 1st toss is H}\Big\} \quad \text{for every } j$$

Then

$$\sum_{n=1}^{\infty} P(A_n) = \sum_{n=1}^{\infty} \frac{1}{2} = \lim_{n\to\infty} \frac{1}{2} \cdot n = \infty$$

$$P(A_n \text{ i.o.}) = P(A_n) = \frac{1}{2} \neq 1$$

as all $\{A_n\}$ are the same.

# Lecture 10: Convergence in Probability and Almost Sure Convergence

Oct 8, 2026

**Def. 5.4 Convergence in Probability**

A sequence of random variables $X_n$ converges in probability to $X$ if for all $\varepsilon > 0$

$$\lim_{n\to\infty} P(|X_n - X| \le \varepsilon) = 1 \qquad \text{or} \qquad \lim_{n\to\infty} P(|X_n - X| > \varepsilon) = 0$$

and is denoted by $X_n \xrightarrow{P} X$.

**Def. 5.5 Almost Sure Convergence**

A sequence of random variables $X_n$ converges almost surely to $X$ if

$$P(\lim_{n\to\infty} X_n = X) = 1$$

and is denoted by $X_n \xrightarrow{a.s.} X$.

**Compare the two:**

$$X_n \xrightarrow{P} X \qquad \lim_{n\to\infty} P(|X_n - X| < \varepsilon) = 1$$
$$X_n \xrightarrow{a.s.} X \qquad P(\lim_{n\to\infty} X_n = X) = 1$$

where

$$\lim_{n\to\infty} X_n = X \Leftrightarrow \left\{\omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\right\}$$
$$P(\lim_{n\to\infty} X_n = X) = P(\left\{\omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\right\}) = 1$$

Written in the same form, for all $\varepsilon > 0$:

$$X_n \xrightarrow{P} X \qquad \lim_{n\to\infty} P(|X_n - X| < \varepsilon) = 1$$
$$X_n \xrightarrow{a.s.} X \qquad P(|X_n(\omega) - X(\omega)| < \varepsilon \text{ a.a.}) = 1$$

"$\xrightarrow{a.s.}$" condition holds almost always: for intersect of the tail.
"$\xrightarrow{P}$" condition satisfied at target.

**Proposition 5.5.1** If for all $\varepsilon > 0$, $P(|X_n - X| > \varepsilon \text{ i.o.}) = 0$, then

$$X_n \xrightarrow{a.s.} X$$

*Proof.*

$$\begin{aligned} P(\lim_{n\to\infty} X_n = X) &= P(\forall \varepsilon > 0,\ |X_n(\omega) - X(\omega)| < \varepsilon \text{ a.a.}) \\ &= 1 - P(|X_n(\omega) - X(\omega)| \ge \varepsilon \text{ i.o.}) \qquad \text{(Corollary 5.2.1)} \\ &= 1 - 0 = 1 \end{aligned}$$

□

**Proposition 5.5.2** If for all $\varepsilon > 0$,

$$\sum_{n=1}^{\infty} P(|X_n(\omega) - X(\omega)| \geq \varepsilon) < \infty$$

then $X_n \xrightarrow{a.s.} X$.

*Proof.* Applying the Borel-Cantelli Lemma. □

**Proposition 5.5.3** If $X_n \xrightarrow{a.s.} X$, then $X_n \xrightarrow{P} X$.
The opposite doesn't hold: $X_n \xrightarrow{P} X \not\Rightarrow X_n \xrightarrow{a.s.} X$.

*Proof.* For any $\varepsilon > 0$, let

$$E_n = \{\omega : \exists m \geq n,\ |X_m(\omega) - X(\omega)| \geq \varepsilon\}$$

Observe that $E_{n+1} \subseteq E_n$, and if $\omega \in \bigcap_{n=1}^{\infty} E_n$ then $X_n(\omega) \overset{a.s.}{\not\longrightarrow} X(\omega)$.

$$\begin{aligned}
\lim_{n\to\infty} P(|X_n(\omega) - X(\omega)| \geq \varepsilon) &\leq \lim_{n\to\infty} P(E_n) \\
&= P\left(\bigcap_{n=1}^{\infty} E_n\right) \\
&\leq P(\{\omega : X_n(\omega) \not\to X(\omega)\}) \\
&= 0 \qquad\qquad \text{since } X_n \xrightarrow{a.s.} X
\end{aligned}$$

So $\lim_{n\to\infty} P(|X_n - X| \geq \varepsilon) = 0$ for every $\varepsilon > 0$, i.e. $X_n \xrightarrow{P} X$. The counterexample for the converse is Example 5.5.4. □

**Example 5.5.4** $X_n$ independent with $P(X_n = 1) = \frac{1}{n}$, $P(X_n = 0) = 1 - \frac{1}{n}$.
For all $\varepsilon > 0$,

$$P(X_n > \varepsilon) = \frac{1}{n} \to 0 \text{ as } n \to \infty$$

So, $X_n \xrightarrow{P} 0$.
However,

$$\begin{aligned}
&P(X_n = 1 \text{ i.o.}) = 1 \\
&\Rightarrow X_n \overset{a.s.}{\not\longrightarrow} 0
\end{aligned}$$

**Theorem 5.5.5** If $X_n \xrightarrow{P} X$, there exists a subsequence such that

$$X_{n_k} \xrightarrow{a.s.} X$$

**Theorem 5.6 Continuous Mapping Theorem**
If $f$ is a continuous function, then

i) $X_n \xrightarrow{a.s.} X \Rightarrow f(X_n) \xrightarrow{a.s.} f(X)$

ii) $X_n \xrightarrow{P} X \Rightarrow f(X_n) \xrightarrow{P} f(X)$

*Proof.* i) $f$ is continuous, it means

$$X_n(\omega) \to X(\omega) \Rightarrow f(X_n(\omega)) \to f(X(\omega))$$

ii) For all $\varepsilon > 0$, there exists a $\delta > 0$, such that $|X_n(\omega) - X(\omega)| \leq \delta$ implies

$$|f(X_n(\omega)) - f(X(\omega))| \leq \varepsilon$$

$$\Rightarrow 1 = \lim_{n\to\infty} P(|X_n(\omega) - X(\omega)| \leq \delta) \leq \lim_{n\to\infty} P(|f(X_n(\omega)) - f(X(\omega))| \leq \varepsilon) \leq 1$$

□

# Lecture 11:

Oct 13, 2026

**Theorem 5.7 Bounded Convergence Thm**
Suppose that $|X_n| \leq M$ a.s. and $X_n \xrightarrow{P} X$,
Then $EX = \lim_{n\to\infty} EX_n$
In general

$$\lim_{n\to\infty} EX_n = E \lim_{n\to\infty} X_n$$

This may NOT be true!
However, under certain conditions, you can switch taking limit and expectation.

$$\begin{cases} \text{Bounded Convergence Theorem} & \text{(BCT)} \\ \text{Monotone Convergence Theorem} & \text{(MCT)} \\ \text{Dominated Convergence Theorem} & \text{(DCT)} \end{cases}$$

**Example 5.6.1 Counter Example**
Suppose the sequence of R.V. $\{X_n\}$

$$\begin{cases} n & \text{with probability } \frac{1}{n} \\ 0 & \text{with probability } 1 - \frac{1}{n} \end{cases}$$

$$\lim_{n\to\infty} X_n = 0 \quad \text{a.s.}$$

$$\Rightarrow E\left(\lim_{n\to\infty} X_n\right) = E(0) = 0$$

However, for each $n \quad E(X_n) = n \times \frac{1}{n} = 1$

$$\lim_{n\to\infty} E(X_n) = \lim_{n\to\infty} 1 = 1$$

The problem is, even though the probability of $X_n$ taking non-zero values is shrinkaging to 0, the value it may take, however, goes sufficiently large.
However, under BCT, the value that $X_n$ may take is boudned, and it can be no longer sufficiently large to offset the convergence of probability, therefore we can have limit of expectation = expectation of limit.

*Proof.* Fix $\varepsilon > 0$ and define

$$\mathcal{G}_n = \{|X_n - X| > \varepsilon\}$$

as $X_n \xrightarrow{P} X \qquad P(\mathcal{G}_n) \to 0$

$$|EX_n - EX| = |E(X_n - X)| \qquad f(x) = |X| \text{ convex}$$

$$\leq E\,|X_n - X|$$

$$= \underbrace{E[|X_n - X|\, I_{\mathcal{G}_n}]}_{\leq 2M \cdot P(\mathcal{G}_n)} + \underbrace{E[|X_n - X| \cdot I_{\mathcal{G}_n^c}]}_{\leq \varepsilon[1-P(\mathcal{G}_n)]}$$

$$\leq \underbrace{2M \cdot P(\mathcal{G}_n)}_{\substack{\to 0 \\ \text{as } n\to\infty}} + \varepsilon \cdot [1 - P(\mathcal{G}_n)]$$

$$\uparrow$$

arbitrary

$$= \varepsilon + P(\mathcal{G}_n)\,|2M - \varepsilon|$$

As this value can be arbitrarily small, we show that $E\,|X_n - X|$ can be arbitrarily small as $n \to \infty$. Note: for BCT, we do not require $X_n \xrightarrow{a.s.} X$, but $X_n \xrightarrow{P} X$ is sufficient. □

**Theorem 5.8 Fatou's Lemma**
If $X_n \geq 0$ a.s. for all $n$, then

$$\liminf_{n\to\infty} EX_n \geq E(\liminf_{n\to\infty} X_n)$$

Proof not required!!!!

**Theorem 5.8.1 Monotone Convergence Theorem**
If $X_n \geq 0$ a.s. and

$$X_n \uparrow X \quad \text{a.s.}$$

$$\uparrow$$

monotonically increasing

then

$$\lim_{n\to\infty} EX_n = EX$$

$X_n \uparrow X$ means for each $\omega \in \Omega$,

$$X_1(\omega) \leq X_2(\omega) \leq \cdots \leq X_n(\omega) \leq \cdots$$

$$\lim_{n\to\infty} X_n(\omega) = X(\omega)$$

*Proof.* Since $EX_n \leq EX$ as $X_n \leq X$ a.s.,

$$\limsup_{n\to\infty} EX_n \leq \lim_{n\to\infty} EX = EX = E \lim_{n\to\infty} X_n$$

However, by Fatou's Lemma, as $X_n \xrightarrow{a.s.} X$ (so $\liminf_{n\to\infty} X_n = X$ a.s.),

$$EX = E(\liminf_{n\to\infty} X_n) \leq \liminf_{n\to\infty} EX_n$$

Therefore

$$\limsup_{n\to\infty} EX_n \leq EX \leq \liminf_{n\to\infty} EX_n$$

$$\Rightarrow \lim_{n\to\infty} EX_n = EX$$

□

**Theorem 5.9 Dominated Convergence Theorem**
If $X_n \xrightarrow{a.s.} X$ and $|X_n| \leq Y$ a.s. for some integrable $Y$, then

$$\lim_{n\to\infty} E(X_n) = E(\lim_{n\to\infty} X_n)$$